

Electrical Collector Systems – Design Standard

Engineering Standard, Revision 2

Sofie Lindqvist · 30 September 2025 · WE-ENG-STD-004

1. Scope

Requirements for medium-voltage collector systems (33 kV onshore, 66 kV offshore), substations and the grid interface. Prepared by Mei Tanaka (Senior Electrical Engineer), approved by Sofie Lindqvist.

2. Voltage levels and topology

Application	Collector voltage	Topology	Max. turbines per string
Offshore wind	66 kV	Radial strings, optional end-loop between adjacent strings	6 × 12–15 MW
Onshore wind	33 kV	Radial	6–8 × 4–6 MW
Solar PV	33 kV	Radial from inverter stations	By cable rating
BESS	33 kV	Radial, one feeder per 25 MW block	–

3. Cable design

- Rating calculated to IEC 60287 with measured soil thermal resistivity; default 1.2 K·m/W only at concept stage.
- Offshore array cables: burial target depth 1.0–1.5 m, cable burial risk assessment mandatory.
- Losses evaluated over the full wind speed distribution, not at rated power only.
- Fibre optic cores in every array and export cable for SCADA and distributed temperature sensing.

4. Protection and earthing

Each string breaker shall have directional earth fault protection. Neutral earthing at the main transformer via resistor limiting earth fault current to 1 kA on 33 kV and 2 kA on 66 kV systems.

5. Grid code compliance

Reactive power capability of 0.95 lagging to 0.95 leading at the point of connection is the design minimum; grid-forming capability shall be evaluated for every new BESS project (see the tech watch note on grid-forming inverters).